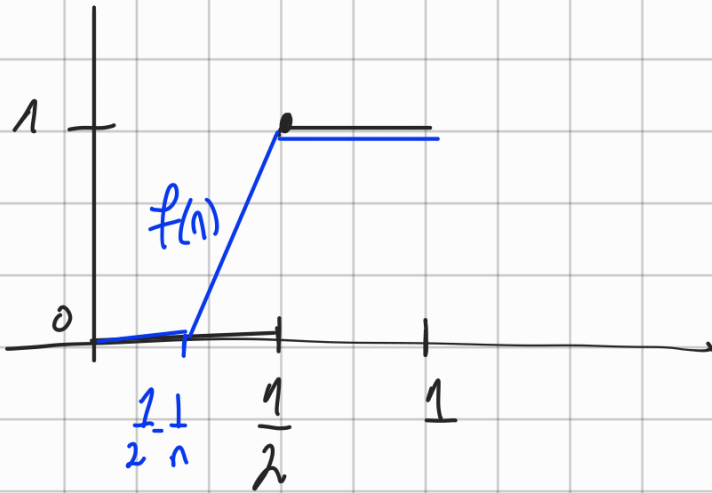


•) $(C[0,1], d_1)_{\mathbb{C}}$ es completo? (veamos que no)



$$g(x) = \begin{cases} 0 & \text{si } x < 1/2 \\ 1 & \text{si } x \geq 1/2 \end{cases}$$

$$\text{Sea } f_n(x) = \begin{cases} 0 & \text{si } x \in [0, 1/2 - 1/n) \\ n(x - (1/2 - 1/n)) & \text{si } x \in [1/2 - 1/n, 1/2) \\ 1 & \text{si } x \in [1/2, 1] \end{cases}$$

1) que $(f_n)_{n \in \mathbb{N}}$ es de Cauchy

Sea $\epsilon > 0$.

$$d_1(f_n(x), f_m(x)) = \int_0^1 |f_n(x) - f_m(x)| dx \leq \frac{1}{2n}$$

$$\frac{1}{2 \min\{n, m\}} < \epsilon$$



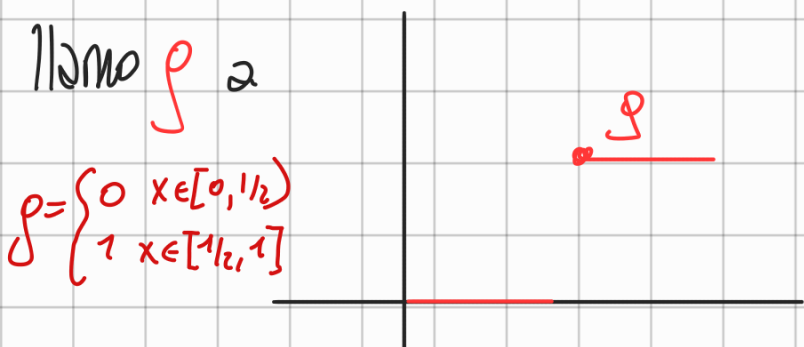
$$\frac{b \cdot h}{2} = \frac{1}{2n}$$

$$\text{Sea } n_0 / \frac{1}{2n_0} < \varepsilon$$

$$\Rightarrow d_1(f_n, f_m) < \varepsilon \quad \forall n, m \geq n_0$$

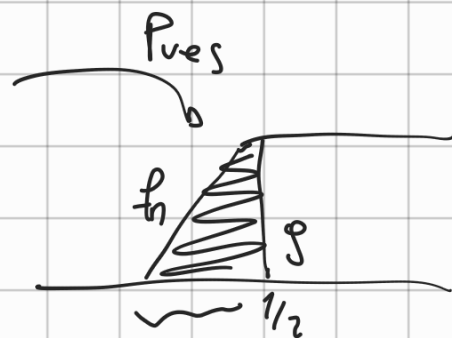
$\Rightarrow f_n$ es de Cauchy

Supongamos $f_n \rightarrow f \in C[0,1]$



$$\int_0^1 |f-g| = \int_0^{1/2} |f-g| + \int_{1/2}^1 |f-g|$$

$$\int_0^1 |f-g| dx \leq \underbrace{\int_0^1 |f-f_n| dx}_{< \varepsilon/2} + \underbrace{\int_0^1 |f_n-g| dx}_{\leq \frac{1}{2n} < \frac{\varepsilon}{2}}$$



$$\Rightarrow \int_0^1 |f-g| dx = 0$$

$$\Rightarrow \int_0^{1/2} |f-g| dx = 0 \quad \gamma \quad f-g \text{ es cont en } [0, 1/2]$$

$$\Rightarrow f=g \text{ en } [0, 1/2]$$

$$\text{y } f=g \text{ en } [1/2, 1]$$

Abs! (f_n no converge)

$$f_n \not\rightarrow f$$

$(C[0,1], d_\infty)$ es completo

Sea $(f_n)_{n \in \mathbb{N}}$ de Cauchy

Sea $x \in [0,1]$. Sea $\varepsilon > 0 \exists n_0 \in \mathbb{N} / \forall n, m \geq n_0$

$$\sup_{x \in [0,1]} |f_n(x) - f_m(x)| < \varepsilon$$

Si x es fijo $(f_n(x))_n \in \mathbb{R}$

$$f(x) = \lim_{n \rightarrow \infty} f_n(x)$$

Paso 1) Definir f candidato a límite

2) $d_\infty(f_n, f) \xrightarrow{n} 0 \Rightarrow f_n$ converge a f

3) Ver que f es continua

Sea n fijo

Sea $n, m \geq n_0$ tal que $\sup_{x \in [0,1]} |f_n(x) - f_m(x)| < \varepsilon$

f_n es de Cauchy

(n es fijo)

$$\lim_{m \rightarrow \infty} |f_n(x) - f_m(x)| < \varepsilon$$

→ ¿? si no lo crees hace $\geq \varepsilon$ y llegas a un abs!

||

$$\forall x \in [0,1] |f_n(x) - f(x)| < \varepsilon$$

3) ejercicios